

Naive Bayes

Step – 1: Read the data

Step – 2: Preprocess or EDA

Step – 3: Model development

- *Machine learning*
 - *Linear regression*
 - ❖ $y = b_0 + b_1 * x_1 + \dots$
 - *Logistic regression*
 - ❖ $\text{sigmoid} = \frac{1}{1+e^{-x}}$
 - *Decision Tree*
 - ❖ *Entropy :*
 - ❖ *Gini :*
- *Deep learning*
- *NLP*
- *GenAI*

Step – 4: Model evaluation

Step – 5: Hyper parameter tuning the model

Step – 6: Model deployment

*Naive Bayes is a classification algorithm based on **Bayes theorem***

It involves probability concept

In the probability there are some terminologies

- ***Experiment :***
 - *we will conduct some task to get an output*
 - *For example toss a coin, Roll a die*
- ***Event:***
 - *Outcome from the experiment*
 - *result from the experiment*
 - *When we toss a coin, there are two results are possible*
 - *Getting an head or Getting a tail*
 - ***Event means we are looking for a specific result***
 - *Roll a die and we want 6*
 - *Toss a coin and getting head*
- ***Random variable:***
 - *Variable assigned for the outcome of an event*

- Roll a die what is the probability getting 4 = $\frac{1}{6}$ = assigned to an x
- Where x is a random variable

Experiment: Toss a coin

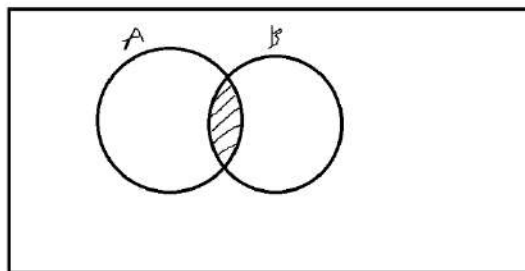
Event: Getting a head

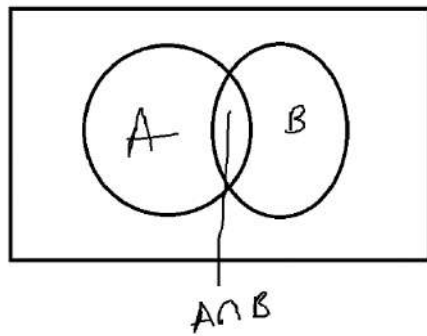
$$RV \ x = \frac{1}{2}$$

- A and B are two events
- $P(A)$ = probability of getting that event A
- $P(B)$ = Probability of getting B
- A = travel from Hyd to Blr
- B = Chance of Rain
- $P(A)$ = Probability to travel from Hyd to Blr
- $P(B)$ = Probability to get a Rain

Addition Theorem:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$





$$A = \{1, 2, 3\}$$

$$B = \{3, 4, 5\}$$

$$\{1, 2, 3, 3, 4, 5\} - \{3\}$$

U: OR $\cap$ = And

SPECIAL case: Mutually Exclusive events

A and B are Mutually Exclusive events means, If A occurs B will not occur

Example – 1:

- A = Getting a Head
- B = Getting a Tail
- **Experiment: Toss a Coin**
- Outcome: Either Head or Tail
- When head appear, Tail will not appear vice versa

Example – 2:

A = Getting a job for Aravind

B = Getting a job for Adithya

Only One Vacancy available

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B)$$

Multiplication Theorem:

$$P(A \cap B) = P\left(\frac{A}{B}\right) * P(B) \text{ or } P\left(\frac{B}{A}\right) * P(A)$$

$$P\left(\frac{A}{B}\right) = \text{Conditional probability}$$

$P(A)$ given B :

- Probability of A when B is available
- chance of Occurrence of A is depends on B
- Thats why It is called as Conditional Probability
- Dr occurs first, then we will think about Nr
- B will come first then we will think about A

A : Travel From Hyd to BLR

B : Getting a Rain

$P(A)$: Chance of travel Hyd to BLR

$P(B)$: Chance of Getting Rain

$P\left(\frac{A}{B}\right)$: Chance of travel Hyd to BLR , when rain occurs

First rain is Coming, Then we are thinking about Go to blr

$P\left(\frac{B}{A}\right)$: Chance of getting rain, when we are tarvel from Hyd to BLR

First we are traveling , Whil Traveling we are thinking about Rain

A : Chance of Pass exam

B :Tougness of Question Paper create $P(A)$ $P(B)$ $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$

$P(A)$: Chance of Pass Exam

$P(B)$: Chance of toughness of Qp

$P\left(\frac{A}{B}\right)$: Chance of pass exam when QP is tough

$P\left(\frac{B}{A}\right)$: Chance of toughness of exam when he pass the exam

$X = \text{input data}$

$y = \text{output data}$

$P\left(\frac{X}{Y}\right) \text{ or } P\left(\frac{Y}{X}\right)$

Once model is Developed , Model will take the data (x) based on data

It will give output (y)

so we required $p\left(\frac{y}{x}\right)$

Special case:

Independent Events

Independent events means , Two events are not related each other

assume that Your qn paper and pass the exam not at all related

You are travelling to blr and rain not at all related

$$P\left(\frac{A}{B}\right) = P(A)$$

$$P\left(\frac{B}{A}\right) = P(B)$$

$$P(A \cap B) = P(A) * P(B)$$

1) Addition theorem: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

2) Mutually Exclusive events = $P(A \cup B) = P(A) + P(B)$

3) Multiplication theorem = $P(A \cap B) = P\left(\frac{A}{B}\right) * P(B) \text{ or } P\left(\frac{B}{A}\right) * P(A)$

3) Independent Events: $P(A \cap B) = P(A) * P(B)$